Kiara College

Year 12 Semester Two Examination, 2015

Question/Answer Booklet

CHEMISTRY
Stage 3

Student Name: ______________________

Teacher Name: ______________________

Time allowed for this paper

Reading time before commencing work: ten minutes
Working time for paper: three hours

Section	Marks
1	/25
	/50
2	**/70**
3	**/80**
total	**/200**
	%

Materials required/recommended for this paper

To be provided by the supervisor

This Question/Answer Booklet
Multiple-choice Answer Sheet
Chemistry Data Sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction tape/fluid, eraser, ruler, highlighters

Special items: up to three non-programmable calculators approved for use in the WACE examinations

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised notes or other items of a non-personal nature in the examination room. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Suggested working time (minutes)	Marks available	Percentage of exam
Section One: Multiple-choice	25	25	50	25	25
Section Two: Short answer	11	11	60	70	35
Section Three: Extended answer	5	5	70	80	40
				Total	100

Instructions to candidates

1. Answer the questions according to the following instructions.

   Section One: Answer **all** questions on the separate Multiple-choice Answer Sheet provided. For each question, shade the box to indicate your answer. Use only a blue or black pen to shade the boxes. If you make a mistake, place a cross through that square, then shade your new answer. Do not erase or use correction fluid/tape. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

   Sections Two and Three: Write answers in this Question/Answer Booklet.

3. When calculating numerical answers, show your working or reasoning clearly. Express numerical answers to **three significant figures and include appropriate units** where applicable.

4. You must be careful to confine your responses to the specific questions asked and to follow any instructions that are specific to a particular question.

5. Spare pages are included at the end of this booklet. They can be used for planning your responses and/or as additional space if required to continue an answer.
   - Planning: If you use the spare pages for planning, indicate this clearly at the top of the page.
   - Continuing an answer: If you need to use the space to continue an answer, indicate in the original answer space where the answer is continued, i.e. give the page number. Fill in the number of the question that you are continuing to answer at the top of the page.

Section One: Multiple-choice 25% (25 Marks)

This section has **25** questions. Answer **all** questions on the separate Multiple-choice Answer Sheet provided. For each question shade the box to indicate your answer. Use only a blue or black pen to shade the boxes. If you make a mistake, place a cross through that square, do not erase or use correction fluid, and shade your new answer. Marks will not be deducted for incorrect answers. No marks will be given if more than one answer is completed for any question.

Suggested working time: 50 minutes.

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1. Which one of the following elements has the highest electronegativity?

   (a) B
   (b) Be
   (c) Ba
   (d) Bi

2. Which one of the following elements has the shortest atomic radius?

   (a) N
   (b) Ne
   (c) Na
   (d) Ni

3. Which one of the following pairs of solutions will **not** form a white precipitate when mixed together?

   (a) sodium carbonate and magnesium chloride
   (b) ammonium carbonate and zinc chloride
   (c) copper(II) sulfate, and barium nitrate
   (d) sodium iodide and silver nitrate

4. Which one of the entries in the table below has correctly placed the following substances, regarding their boiling points and solubility in water?

Substances: ethanoic acid, butanoic acid, ethanol, butanol

	Highest boiling point	Lowest solubility in water
(a)	butanoic acid	ethanol
(b)	butanoic acid	butanol
(c)	butanol	ethanoic acid
(d)	butanol	butanoic acid

5. Examine the first five ionisation energies of element **A** below.

	1^{st}	2^{nd}	3^{rd}	4^{th}	5^{th}
Ionisation Energies ($kJ\ mol^{-1}$)	793	905	3 392	4 167	5 111

Which one of the following is the most likely formula of the oxide of element **A**?

(a) A_2O_3
(b) AO_2
(c) A_2O
(d) AO

6. Which of the following will conduct an electric current?

i molten sulfur
ii a saturated solution of silver sulfate
iii solid silver
iv solid silver sulfide

(a) i and ii only
(b) ii and iii only
(c) i and iv only
(d) i, ii and iv only

7. In which one of the following substances are dispersion forces the most significant type of intermolecular force?

(a) solid carbon dioxide
(b) liquid ethanol
(c) solid butanoic acid
(d) solid sodium chloride

8. Which one of the following species contains lone (non-bonding) pairs of valence electrons?

(a) C_2H_4
(b) NH_4^+
(c) H_2S
(d) CH_4

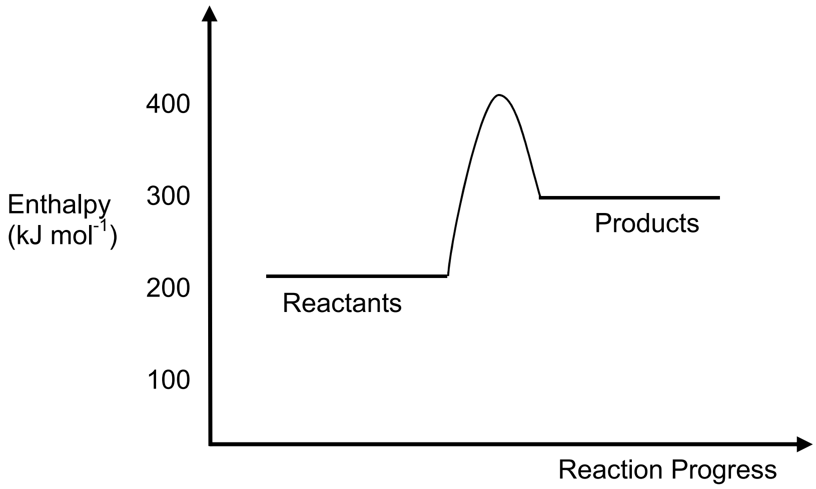
9. Which one of the following substances is likely to be the most soluble in water?

(a) HF
(b) H_2S
(c) H_2
(d) CH_4

10. If 2.00 mg of sodium chloride crystals were dissolved in distilled water and the solution made up to 100.0 mL, which one of the following would most correctly describe the solution? Assume density of the resulting solution to be 1.00 gmL^{-1}.

(a) A 2.00% salt solution
(b) A 20.0 gL^{-1} salt solution
(c) A 0.342 $molL^{-1}$ salt solution
(d) A 20.0 ppm salt solution

11. An energy profile diagram for a chemical reaction is shown below.



Estimate the activation energy for the **reverse** reaction.

(a) + 400 kJ mol^{-1}
(b) + 200 kJ mol^{-1}
(c) - 200 kJ mol^{-1}
(d) + 120 kJ mol^{-1}

12. Which one of the following correctly arranges 0.01 mol L^{-1} solutions of the substances in the order of decreasing pH, from highest to lowest?

(a)	$Ba(OH)_2$	$NaOH$	Na_2CO_3	$NaNO_3$	NH_4NO_3
(b)	$NaOH$	$Ba(OH)_2$	Na_2CO_3	NH_4NO_3	$NaNO_3$
(c)	$Ba(OH)_2$	$NaOH$	$NaNO_3$	Na_2CO_3	NH_4NO_3
(d)	$NaOH$	$Ba(OH)_2$	Na_2CO_3	$NaNO_3$	NH_4NO_3

13. Consider the equilibrium system below:

$$H_2O_{(\ell)} + NH_{3(aq)} \rightleftharpoons OH^-_{(aq)} + NH_4^+{}_{(aq)}$$

Which one of the following statements is **false**?

(a) NH_3 is the conjugate base of NH_4^+.
(b) The water is acting as a base.
(c) Addition of water will favour the forward reaction.
(d) The system can oppose an increase in pH by favouring the reverse reaction.

14. Which one of the following 1.0 mol L^{-1} solutions will have the lowest pH?

(a) $HC\ell(aq)$
(b) $H_3PO_4(aq)$
(c) $H_2SO_4(aq)$
(d) $NH_4C\ell(aq)$

15. Which one of the following species listed below contains sulfur with the highest oxidation state?

(a) H_2SO_3
(b) S_8
(c) SO_2
(d) $MgSO_4$

16. In which one of the following reactions is oxygen undergoing disproportionation (being oxidised and reduced)?

(a) $2\ C\ell O^- + 4\ H^+ \rightarrow C\ell_2 + 2\ C\ell^- + 2\ H_2O$
(b) $2\ H_2O_2 \rightarrow O_2 + 2\ H_2O$
(c) $5\ MnO_2 + 4\ H^+ \rightarrow 2\ MnO_4^- + 3\ Mn^{2+} + 2\ H_2O$
(d) $2\ FeO + 3\ CO_2 \rightarrow Fe_2O_3 + 3\ CO$

17. By referring to the table of standard electrode potentials on the Chemistry Data Sheet, predict which one of the following pairs of substances will undergo a chemical reaction.

(a) Iron(III) nitrate solution and sodium chloride
(b) Sodium chloride solution and solid iodine
(c) Gaseous chlorine and solid silver
(d) Hydrogen peroxide solution and lead(II) sulfate solution

18. In a class experiment, Gowtham was given three unknown metals labelled X, Y, and Z, and their corresponding 1.0 $molL^{-1}$ nitrate solutions. He also had access to a standard hydrogen gas/hydrogen ion half-cell, leads, a salt bridge and a voltmeter. He was able to set up and measure the voltage generated using different combinations of the four half-cells. His results are recorded in the table below. The room conditions were a temperature of $25^{o}C$, and 1.0atm of pressure.

Anode	Cathode	Voltage (Volts)
Z	Y	2.48
X	Y	1.20
X	H^+/H_2	0.40

What would the chemical formula of the metal Z?

(a) Ag
(b) Cd
(c) Mn
(d) Aℓ

19. Which one of the following has a different empirical formula to the other three?

(a) butanoic acid
(b) methyl propanoate
(c) ethanal
(d) propyl propanoate

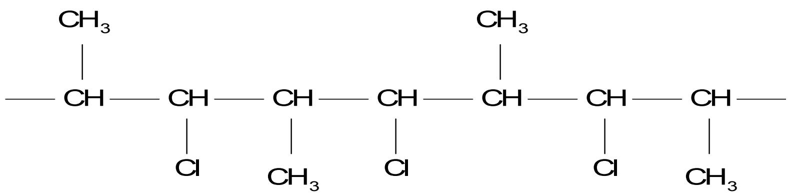
20. Which one of the following pairs of compounds would form propyl methanoate when warmed with concentrated sulfuric acid?

(a) CH_3OH and CH_3CH_2COOH
(b) $CH_3CH_2CH_2OH$ and CH_3OH
(c) $HCOOH$ and $CH_3CH_2CH_2OH$
(d) $CH_3CH_2CH_3$ and $HCOOH$

21. Which one of the following is an addition reaction?

(a) $CH_3CH_2CH_2CH_2CH_3$ + $Br_2 \rightarrow$ $CH_3CHBrCH_2CH_2CH_3$ + HBr
(b) $CH_3CHCHCH_2CH_3$ + $H_2 \rightarrow$ $CH_3(CH_2)_3CH_3$
(c) C_6H_6 + $CH_3C\ell \rightarrow$ $C_6H_5CH_3$ + $HC\ell$
(d) n $HOCH_2CH_2OH$ + n HOOCCOOH $\rightarrow$ $(-OCH_2CH_2OCOCOO-)_n$ + 2n H_2O

22. Examine the section of the polymer shown below.



Which one of the following monomers could be used to make this polymer?

(a) 1-methyl-2-chloroethene
(b) 1-chloroprop-2-ene
(c) 2-chloropropene
(d) cis-1-chloropropene

23. Which one of the following is the correct half-equation for the oxidation of propan-1-ol to propanoic acid?

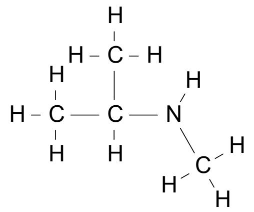
(a) $CH_3CH_2CH_2OH(aq)$ + $H_2O(\ell) \rightarrow$ $CH_3CH_2COOH(aq)$ + 2 $H^+(aq)$ + 2 e^-
(b) $CH_3CH_2CH_2OH(aq) \rightarrow$ $CH_3CH_2CHO(aq)$ + 2 $H^+(aq)$ + 2 e^-
(c) $CH_3CH_2CH_2OH(aq)$ + $H_2O(\ell) \rightarrow$ $CH_3CH_2COOH(aq)$ + 4 $H^+(aq)$ + 4 e^-
(d) $CH_3CH_2CH_2OH(aq)$ + $O_2(g) \rightarrow$ $CH_3CH_2COOH(aq)$ + $H_2O(\ell)$

24. Which one of the following is an α(alpha)-amino acid?

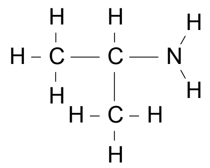
(a) $CH_3CNH_2COOCH_3$
(b) CH_3CNH_2COOH
(c) $NH_2CH_2CH_2COOH$
(d) CH_3CONH_2

25. Which one of the following is a primary amine?

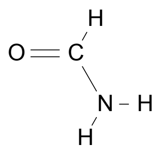
(a)



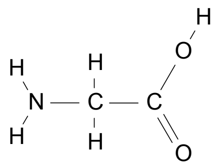
(b)



(c)



(d)



End of Section One